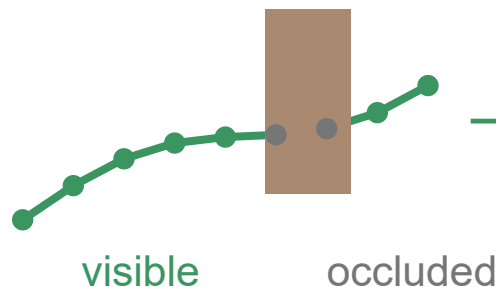


Phase- and visibility-aware structural guidance

Expected structure (3D prior)

C: projected curve



screen overlap
+ nearer utensil
visibility V

Not reconstructed metric 3D

Image evidence (independent)



RGB

Color observer *
current diagnostic

SAM3 *
planned validation

P noodle

P utensil

Phase gate g

approach: OFF
contact / lift
hold: ON

Differentiable surrogate

Weakest-link path
on visible samples

+

Endpoint + two tips
expected contact sites

Validation gate

synthetic masks: pass
RGB semantics: fail

Latent guidance in VACE: not executed

